Expt 6	**Finite Word length Effects**
Date:	**6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB**

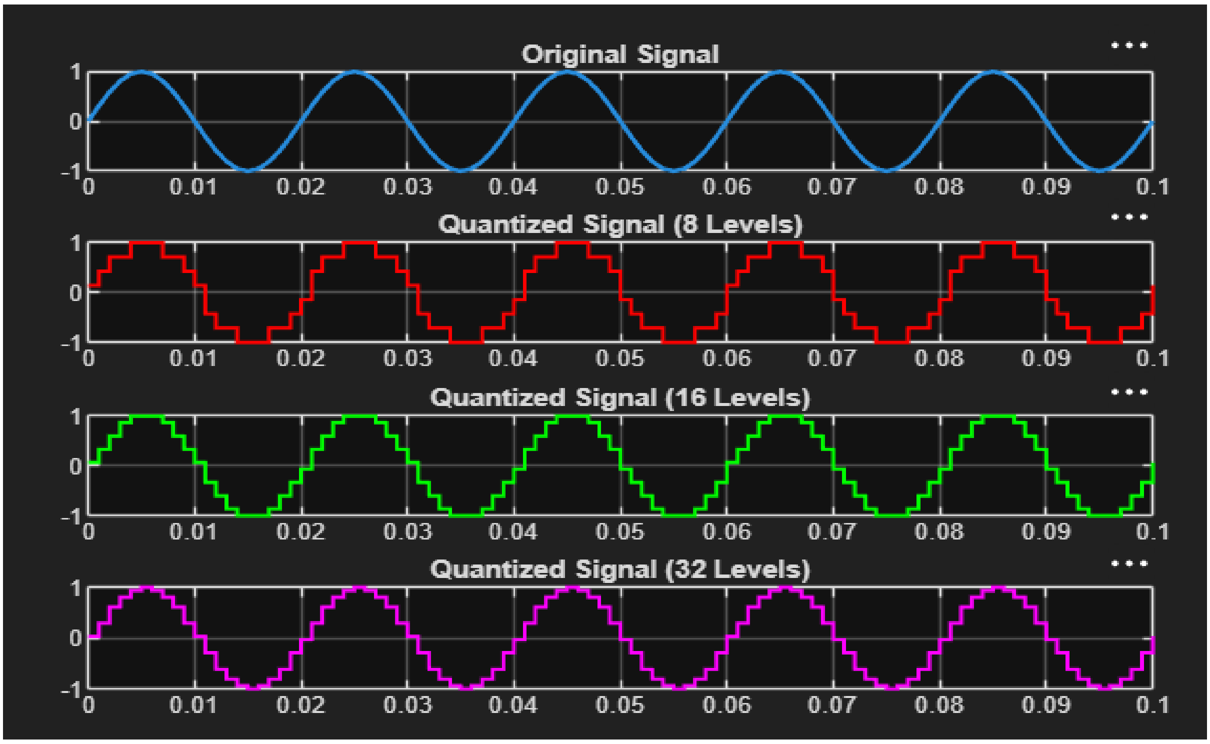
Code:

```
clc;
clear;
close all;

%% Signal
fs = 1000;
f = 50;
t = 0:1/fs:0.1;
x = sin(2*pi*f*t);

%% Quantization Levels
L = [8 16 32];
colors = {'r','g','m'};
figure;

for i = 1:length(L)
    % --- Quantization ---
    % Normalize x to [0,1], scale to levels, round, then map back to [-1,1]
    xq{i} = round((x+1)/2 * (L(i)-1)) * (2/(L(i)-1)) - 1;

    % --- Error & MSE ---
    e{i} = x - xq{i};
    mse(i) = mean(e{i}.^2);
```

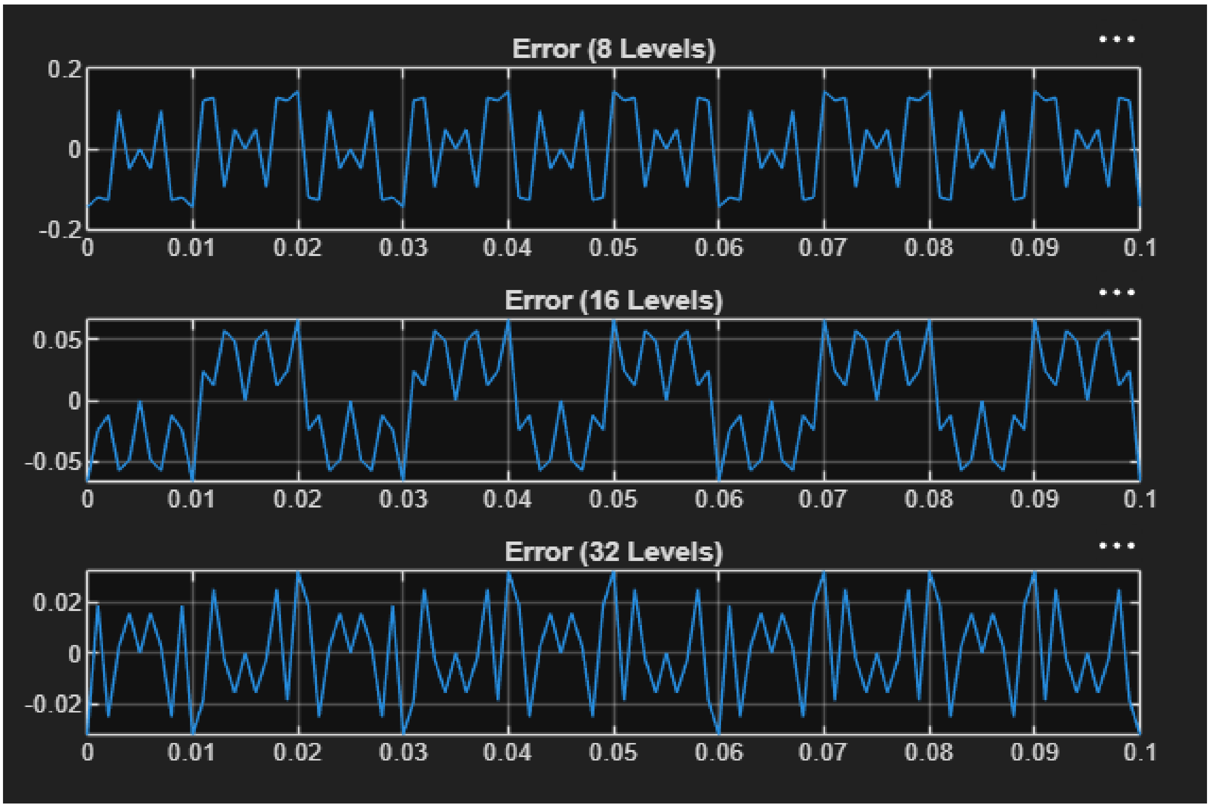
```
    % --- Plot Quantized Signals ---
    subplot(4,1,i+1);
    stairs(t, xq{i}, colors{i}, 'LineWidth', 1.2);
    title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
    grid on;
end
% --- Original Signal ---
subplot(4,1,1);
plot(t, x, 'LineWidth', 1.5);
title('Original Signal');
grid on;
% --- Display MSE ---
fprintf('MSE for 8 Levels  = %f\n', mse(1));
fprintf('MSE for 16 Levels = %f\n', mse(2));
fprintf('MSE for 32 Levels = %f\n', mse(3));

%% Error Plots
figure;
for i = 1:3
    subplot(3,1,i);
    plot(t, e{i});
    title(['Error (' num2str(L(i)) ' Levels)']);
    grid on;
end
```

Output:

Objective 1:


Original Signal
Quantized Signal (8 Levels)
Quantized Signal (16 Levels)
Quantized Signal (32 Levels)
1
0
-1
0 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08 0.09 0.1

Objective 2:


Error (8 Levels)
0.2
0
-0.2
Error (16 Levels)
0.05
0
-0.05
Error (32 Levels)
0.02
0
-0.02
0 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08 0.09 0.1

Expt 6	**Finite Word length Effects**
Date:	**6.2 Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation and Rounding Techniques in MATLAB**

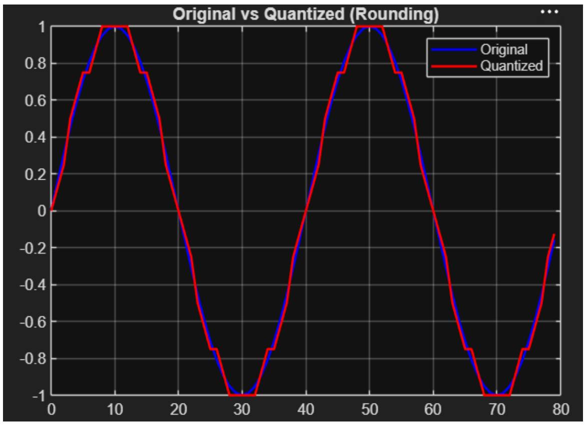
Code:

```
clc;
clear;
close all;

%% Signal
n = 0:79;
x = sin(2*pi*n/40);

%% Quantization
q = 8;
xt = floor(x*q)/q;   % Truncation
xr = round(x*q)/q;   % Rounding

%% Errors
et = x - xt;
er = x - xr;

%% Fig 1: Original vs Quantized (Rounding)
figure;
plot(n, x, 'b', n, xr, 'r', 'LineWidth', 1.5);
grid on;
title('Original vs Quantized (Rounding)');
legend('Original','Quantized');
```

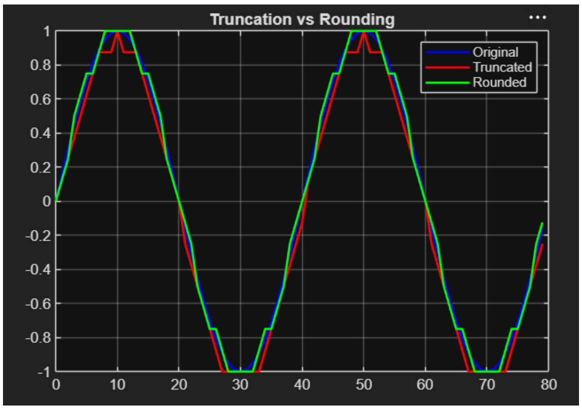
```
%% Fig 2: Truncation vs Rounding
figure;
plot(n, x, 'b', n, xt, 'r', n, xr, 'g', 'LineWidth', 1.5);
grid on;
title('Truncation vs Rounding');
legend('Original','Truncated','Rounded');

%% Fig 3: Quantization Error
figure;
stem(n, et, 'r', 'filled');
hold on;
stem(n, er, 'g', 'filled');
grid on;
title('Quantization Error');
legend('Truncation','Rounding');

%% Fig 4: MSE Comparison
figure;
bar([mean(et.^2) mean(er.^2)]);
set(gca, 'XTickLabel', {'Truncation','Rounding'});
grid on;
title('MSE Comparison');
ylabel('MSE');
```

Output:

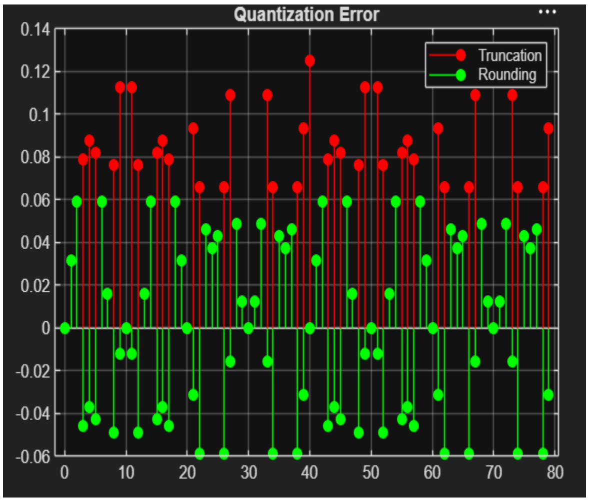
Objective 1:



Objective 2:



Objective 3:



Objective 4: